PRAGATI: Journal of Indian Economy
Vol. 5(2), Jul-Dec 2018, pp. 1-15
DOI: 10.17492/pragati.v5i2.14372
www.journalpressindia.com/pjie
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Macroeconomic Dynamics of Foreign Direct Investment in India: An Empirical Analysis

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ABSTRACT

For the past few years, Foreign Direct Investment (FDI) has become the indicator for Economic Growth, especially in emerging economies. This paper empirically investigates the determinants of FDI flows in India by employing the Auto Regressive Distributed Lag (ARDL) model. The result confirm the existence of a long run equilibrium between the FDI and five explanatory variables, namely exchange rate, Wholesale Price Index, Index of Industrial Production, Trade openness and dummy variable (financial crisis). India's Wholesale Price Index, Exchange Rate volatility and Index of Industrial Production have positively influence the flow of FDI in India and Trade Openness is negatively significant for the flow of FDI in India. The coefficient of the Error Correction Term (ECT) is highly significant with expected sign, which confirm the result of bound test for co-integration. The cumulative sum of recursive residual (CUSUM) test is used for measuring the stability of the model.

Keywords: *FDI; ADF; Co-integration; Bound Test; AIC; ARDL; Economic growth.*

1.0 Introduction

The ultimate determinant of Foreign Investment, like the case of almost all other types of investment, is the return which the investment likely to bring in the short or long run. But there are some other specific factors which push as well as pull foreign capital to the other countries. That is why the flow of foreign capital is not uniform in all countries. While the domestic situations of the certain home countries - lack of opportunities, unfavourable climate for investment, low return, low interest rate, political instability etc. - push their capital to go out of the country, the encouraging situation of

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the host countries like political stability, economic stability, health of the host economy, receptive policies of the government including liberalization and above all the possibility of high return pull the foreign capital to certain countries. However, these push and pull factors are not measurable and cannot be studied scientifically as determinants of Foreign Direct Investment (FDI). These factors directly and indirectly create certain measurable factors like inflation, exchange rate, growth rate, trade openness and balance of payments position which influence the flow of foreign direct investment. International capital inflows have both positive as well as negative impact on the health of the recipient economy. On the positive side, these capital inflows raise the level of economic development by augmenting the domestic investment and widen financial intermediation. But these capital inflows also pose several threats to the domestic economic and financial system of the recipient economy like hike in Inflation, Exchange rate volatility, overheating of the economy and possibility of sudden withdrawal.

The government trade abstractions and policies will affect FDI and contribution to the economy especially the GDP growth rate of the economy. To increase the FDI inflows in the country, Indian government allows frequent equity participation to foreign enterprises apart from providing several incentives such as tax concessions, simplification of licensing procedures and de-reserving various industries like drugs, fertilizers, automobile, insurance etc. But due to significant outflow of foreign reserve in the form of remittances of dividends, profits, royalties etc. In 1973 Government of India set up Foreign Investment Board and enacted Foreign Exchange Regulation Act in order to regulate flow of FDI to India. Additionally, Government of India set up Foreign Investment Promotion Board (FIPB) for processing of FDI proposals in India. In Last Budget speech, Finance minister announced to abolish the FIPB and it is abolished now. The main reason for the abolishment is that more than 92 percent if FDI came from automatic route. FIPB offer single window clearance for application of FDI in India under approval route.

Now India is an attractive destination of Foreign Investment. But Government needs to do more to sustain large Foreign Investment inflows. Policy regime is the key factor for driving Foreign Investment flows in a country. There are various strategies followed by the host economy to attract FDI. Liberalization, deregulation, free trade agreement, incentives, tax concessions etc. are the different strategies followed by the host nation to attract FDI. The size of the market is the prime consideration prior to invest money in a foreign country. Beneath liberalization there must be other elements too which promoted FDI in India. This study attempts to trace the reasons of FDI flows in India. In other wards determinants of FDI flows in India.

2.0 Determinants of FDI Inflows: A Review of Literature

To understand comprehensively favourable and unfavourable conditions of FDI and its determinants, it is necessary to make a taxonomic review of related literatures which shades lighter on the concept of FDI and focuses on determinants of FDI.

Wang (2009) and Prasad et.al (2007) discusses the role of FDI inflows to affect host country economic growth. They argues that FDI should have a positive effect on economic growth. There are large number of macro-economic factors affecting the flow of Foreign Direct Investment in India. Foreign Direct Investment flow is mainly depend on the rate of Inflation of the host economy. Akinboade et.al (2001) state that low Inflation indicates internal economic stability in the host nation. A high Inflation rate imply economic instability associated with inappropriate government policies, especially the monetary and fiscal policy mix (Macpherson, 2013). Khan and Mitra (2014) opine that high rates of Inflation distort the economic activities, contributing to lesser inflow of capital. It affects profitability as higher costs can lead to increased costs and lower earnings. Thus, stable Inflation rate is desirable to attract Foreign Capital. So the expected relationship between FDI and Inflation is negative.

Another factor which influence the FDI investment is volatility of Exchange Rate. Affecting the Exchange Rate volatility on Foreign Direct Investment flow is a debatable question. If the FDI's objective is to concentrate the host country market, the appreciation of the host currency attracts the FDI to enter into the host country. On the other hand, if the FDI's objective is for reexport purpose, appreciation of the host currency reduces the FDI inflows through lower competitiveness. According to Banga (2003) Foreign Direct Investment adversely affect the volatility of Exchange Rate. High volatility of Exchange Rate indicates uncertainty regarding the future economic and business aspects of the host country. Ellahi (2011) and Dhakal et.al (2010) also examine the behaviour of Foreign Direct Investment flows in relation to the volatility of Exchange Rate. They support the above view. That means Exchange Rate volatility leads to negative effect on FDI flows.

Index of Industrial Production is usually used as a proxy for measuring growth rates in real sector. IIP is the numeral denoting the status of industrial production during a certain period. Maheswari (2015) identifies that inflow of FDI inflows in the host economy mainly depend upon the strength and trend of Index of Industrial Production. According to Reenu (2015) using Regression Analysis identified market size or IIP as the major determinants of FDI inflows. So the expected relationship of FDI and IIP is positive. Balasubramanyam et.al (1996) also agree with this statement.

Another important factor which influence the flow of FDI is the host country policies and philosophy towards international trade. This is the proportion of trade (imports and exports) to GDP (IIP). Trade openness enhances the attraction of foreign investors to invest in the country. Foreign firms that operate its business in the country could freely trade their commodities and services to other nations. Hence a positive outcome of trade openness on FDI is hypothesized. Zaman, et.al (2012) stated that there is a positive relationship between FDI and trade openness.

3.0 Problem Statement

In most developing countries attracting FDI is a major instrument for promoting economic development. Different countries use diverse terminology, promotion strategies like incentives and tax concessions to attract FDI. Generally developing countries offer incentives to multinational companies (MNCs) for attracting FDI for the development of the economy. These International capital inflows especially FDI have both positive as well as negative impact on the health of the recipient economy. On the positive side, these capital inflows raise the level of economic development. FDI mostly focused on extractive sectors rather than on productive sectors, manufacturing and infrastructural development does not hold any promise of sustainable development for a developing economy. This study is mainly concentrate on what are the key macro-economic variables affecting the performance of FDI for the development of the host economy. Beneath liberalization there must be other factors too which prompted FDI in India. This study attempts to trace the reason for FDI in India. In other wards determinants of FDI in India.

4.0 Objectives of the Study

- To study the structure, composition and trends of Foreign Direct Investments in India in the post reform period.
- To empirically evaluate the explanatory role of macroeconomic variables for affecting Foreign Direct Investment flows in India.

5.0 Research Methodology

5.1 Data source

This study is mainly based on secondary sources. Most of the data are collected from the RBI database, Handbook of statistics on Indian economy, Indian securities

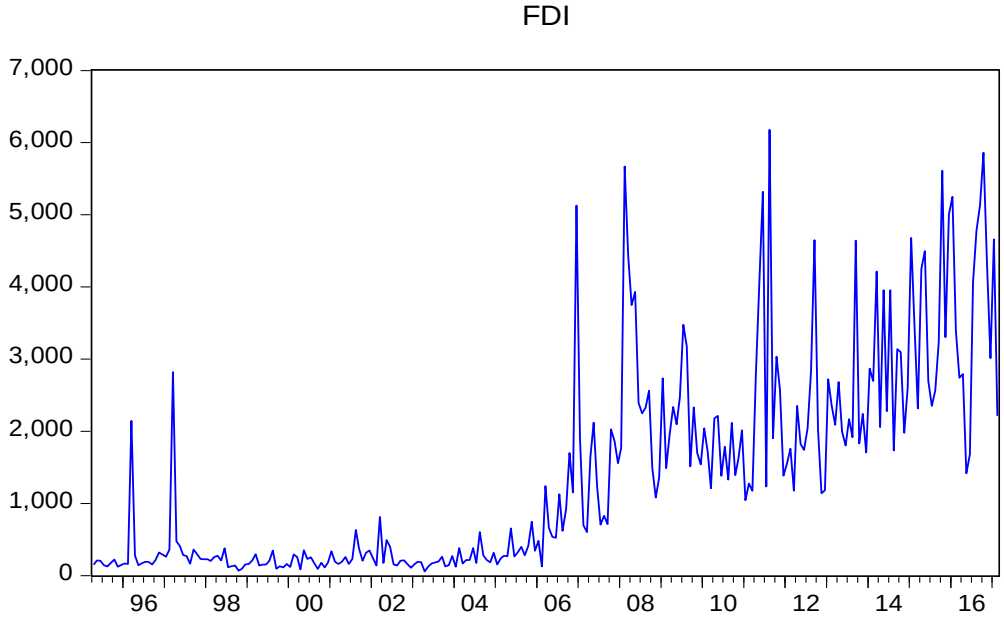
market review, SEBI handbook, SEBI annual report, etc. The monthly data gathered for the study covers from 1995 to 2018.

5.2 Data analysis

Descriptive Statistics have been practiced to know the structural attributes of data. To analyse the determinants of macro-economic indicators of FDI flows in India is analysed through Auto Regressive Distributed Lag (ARDL) model. For checking the stationary properties and unit root of the data series Augmented Dicky Fuller (ADF) test is used. Akaike Information Criteria (AIC) suggest the optimal lag length for ARDL model. CUSUM test is used for measuring the stability of the model.

The growth of FDI has been significant after the economic reforms 1991. More than 90 percent of FDI came from automatic route. Government liberalize the FDI norms and regulations to attract international capital. It will improve the ease of doing business. Latest FDI norms is the clear indication to attract international funds to fuel India's economic growth. Figure 1 reveals in detail trends of FDI inflow in India during the period 1995 to 2018. The average monthly flow of FDI is $ 1370.29 US million.

Figure 1: FDI Inflow in India (US $ Million)

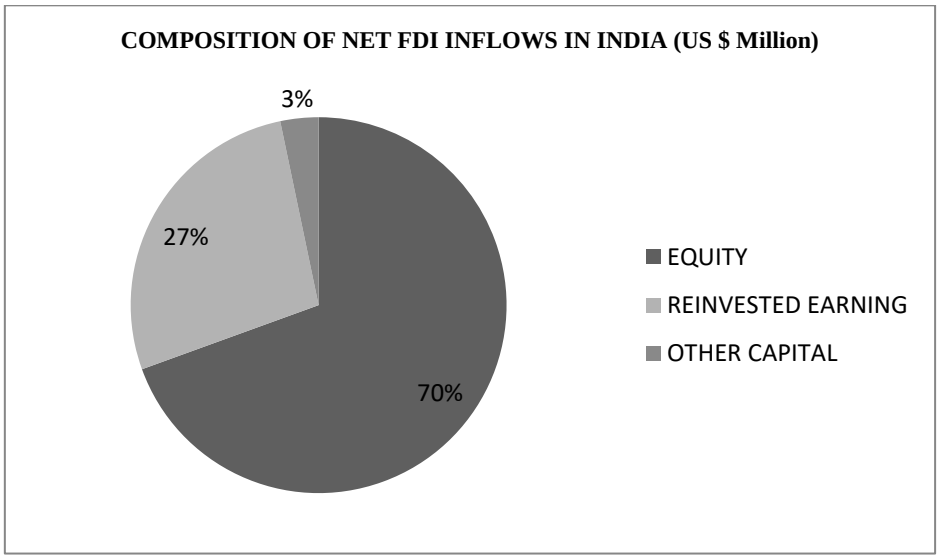


Source: *Compiled from Handbook of Statistics of Indian economy*

6.0 Composition of Foreign Direct Investment

The major bodies constituted for FDI are Department of Industrial Policy and Promotion (DIPP), Secretariat for Industrial Assistance (SIA), Ministry of Commerce and Industry Reserve bank of India (RBI) etc. are the various organisation to keep FDI statistics. Prior to 2000 FDI statistics compiled by the RBI in the Balance of Payments included only equity capital. This contributed to an underestimation of FDI inflows. The accounting system was revised in 2000 in order to line up the FDI data-reporting system with international best practices. FDI has three components, viz., Equity capital[1], Reinvested earnings[2] and Intra-company loans[3]. The major contribution of FDI flows through equity capital. Components of reinvested profits and other capital of total FDI Inflows are comparatively low. Equity capital was 70%, reinvested earnings was 27%, other capital was 3% of the total FDI flows (refer Figure 2). In short, FDI in Equity is a most preferred channel of FDI in India.

Figure 2: Composition of Foreign Direct Investment in India



Source: *Compiled from Handbook of statistics of Indian economy*

7.0 Empirical Model

FDI inflows in India is formulated with the five macroeconomic determinants of independent variables. It includes Economic Growth (IIP), Inflation Rate (WPI),

Exchange Rate (NEER) and Trade Openness (TO) and financial crisis (dummy variable) of host country. The expected relationship between FDI and its macroeconomic determinants in India is exhibits in Table 1.

We develop a linear equation model such that:

FDI = f (NEER, WPI, IIP, TO, DM)

Table 1: Expected Relationship of FDI and Macro-Economic Variables in India

Variable Name	Description	Expected Relationship
NEER	Exchange Rate	Positively or Negatively related
WPI	Wholesale Price Index	Negatively related
IIP	Index of Industrial Production	Positively related
TO	Trade Openness (Government Policy)	Positively related

7.1 Unit root test

The unit root test is used to determine or checking the stationary of the data. Table 2 indicates that the upshot of an ADF test of macroeconomic determinacy of FDI in India. It is evident from the table the order of integration of all the variables used in the subject field is either nil or one i.e. I(0) or I(1). As can be seen from Table 2, Economic Growth (IIP), Inflation Rate (WPI) and Exchange Rate (NEER) and Trade Openness (TO) are integrated of order one, I(1) and FDI is integrated of order I(0).

Table 2: Augmented Dickey Fuller Unit Root Test

Variables	Stationary
FDI	I(0)
NEER	I(1)
WPI	I(1)
IIP	I(1)
TO	I(1)

8.0 ARDL Model

The Auto Regressive Distributed Lag (ARDL) is a technique that allows to simultaneously estimate the short-run and long-run coefficients of macroeconomic determinants of FDI in India. To prove the existence of the long - run relationship among the variables in the model, Pesaran et. al. (2001) introduced Auto Regressive Distributed

Lag (ARDL) model. In ARDL the first step is to investigate the long-run relationship among variables in the model using an F - test. The second step of this analysis is to estimate the coefficients of the long-run relationship with the error correction representation of the ARDL model. Here the dependent variable is always affected by number of number of explanatory variables in the current period as well as in the earlier period.

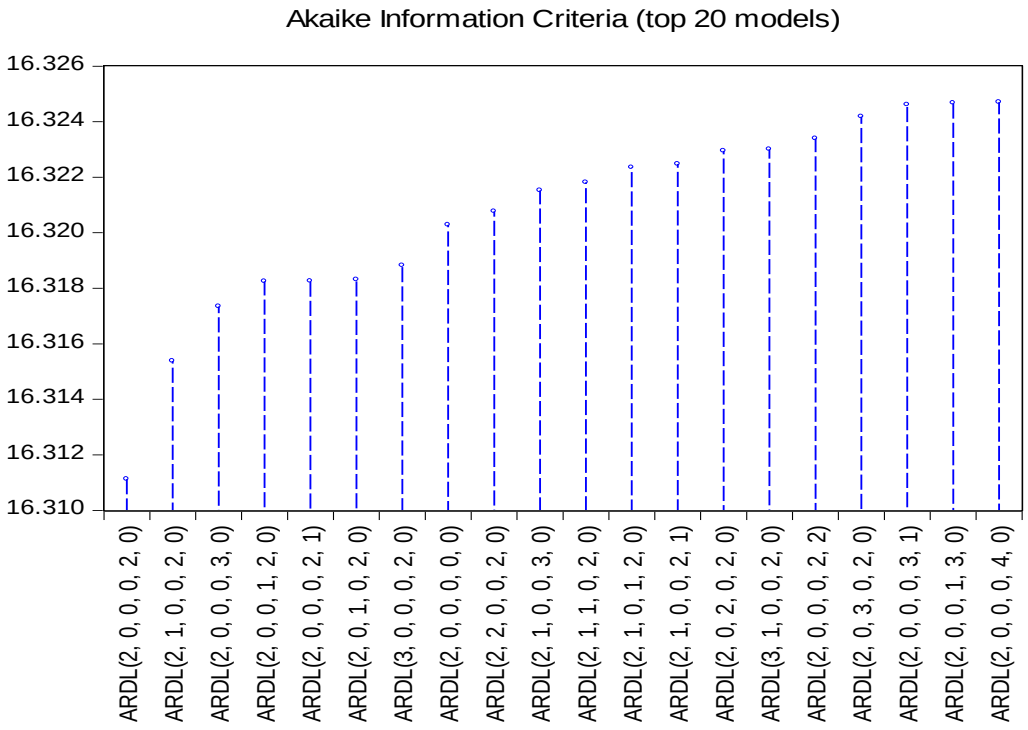
Table 3: ARDL Test - Determinants of FDI in India

Dependent Variable: FDI				
Method: ARDL				
Variable	**Coefficient**	**Std. Error**	**t-Statistic**	**Prob.***
FDI(-1)	0.252752	0.061677	4.098008	0.0001
FDI(-2)	0.202498	0.062881	3.220315	0.0014
NEER	19.77314	11.67924	1.693015	0.0917
WPI	18.50044	9.024865	2.049941	0.0414
IIP	10.63178	4.743283	2.241440	0.0259
TO	0.023514	3.090589	0.007608	0.9939
TO(-1)	4.921512	3.881227	1.268030	0.2060
TO(-2)	-7.703344	3.109982	-2.476974	0.0139
DM	126.5090	249.6979	0.506648	0.6128
C	-4088.961	1841.472	-2.220485	0.0273
R-squared	0.688859	Mean dependent var		1379.429
Adjusted R-squared	0.677702	S.D. dependent var		1450.789
S.E. of regression	823.6311	Akaike info criterion		16.30288
Sum squared resid	1.70E+08	Schwarz criterion		16.43946
Log likelihood	-2117.526	Hannan-Quinn criter.		16.35778
F-statistic	61.74534	Durbin-Watson stat		1.985614
Prob(F-statistic)	0.000000			
*Note: p-values and any subsequent tests do not account for model Selection				

$$FDI_t = \beta_0 + \beta_1 FDI_{t-1} + \beta_2 FDI_{(t-2)} + \beta_3 NEER_t + \beta_4 WPI_t + \beta_5 IIP_t + \beta_6 TO_t + \beta_7 TO_{t-1} + \beta_8 TO_{(t-2)} + \beta_9 DM_t + \epsilon$$

where t-1 is variable's lagged value by one period and t-2 is variable's lagged value by two period, the lag length was determined automatically by Akaike Information Criterion (AIC) (refer Figure 3).

Figure 3: Optimum Lag Criteria of FDI and Macroeconomic Variables in India



9.0 Optimum Lag Length Criteria

Akaike Information Criterion (AIC) used to determine the optimum lag length of the model. That means how many lags should be used for a particular model. In all times the lower the AIC value, the better the model. The empirical analysis reveals that 20 best model with lowest AIC values. As seen in Figure 3, the lowest AIC value shows that the optimum lag length is ARDL (2, 0, 0, 0, 2, and 0).

The Breusch-Godfrey (BG) LM Test (Breusch, 1978) used for testing Serial Correlation of data. The results shows in Table 4 describes P value is greater than 0.05, which indicate that there is no auto correlation. That leads to no problem of Serial Correlation.

RESET Test stands for Regression Specification Error Test *(*Ramsey, 1969) propounded by Ramsey in 1969. This test is mainly focused on model specification. The result indicates that (Table 5) gives the estimated probability value as 0.34 which is

greater than 0.05. This suggests that the model is well specified and without significant omitted variables.

Table 4: Breusch-Godfrey Serial Correlation LM Test

F-statistic	0.205285	Prob. F(2,249)	0.8146
Obs*R-squared	0.429648	Prob. Chi-Square(2)	0.8067

Table 5: Ramsey RESET Test

Ramsey RESET Test				
Omitted Variables: Squares of fitted values				
	Value	df	Probability	
t-statistic	0.948908	250	0.3436	
F-statistic	0.900427	(1, 250)	0.3436	
F-test summary:				
	Sum of Sq.	df	Mean Squares	
Test SSR	611066.6	1	611066.6	
Restricted SSR	1.70E+08	251	678371.6	
Unrestricted SSR	1.70E+08	250	678640.8	

9.1 ARDL bound test approach for co-integration

The primary function of estimating ARDL model is employed for applying bound test. This test is used to investigate the long-run relationship or co-integration among variables in the model. There is no long run relationship between FDI inflows and macro-economic variables in India is the null hypothesis of the study. The result of the bound test are described in Table 6. The computed F-statistic is 10.13 which is more than the upper bound at five percent level. It indicates that there is a long term relationship between variables of the model. That means reject the null hypothesis. So all variables are co-integrated or there is a long run relationship among variables. Therefore there is a co-integration between macro-economic variables and FDI in India.

The Table 7 estimates the result of coefficient of long run relationship between macroeconomic variables and FDI in India by applying ARDL Methodology. The impact of IIP growth contributes positively and statistically significant to FDI flows. A high level of IIP growth (economic growth) is a solid indication of market opportunities. Thus investors are expecting more revenue from commercial enterprises in India. The impact of WPI is positive and statistically significant on the flow of FDI in India. The level of inflation in the economy or price stability has significantly contributed in explaining the flow of FDI in India. This is supported by Anitha (2012) discovered an

unexpected positive relationship between Foreign Direct Investment and Inflation. The impact of Exchange Rate volatility has positively and statistically significant of FDI flows in India. Empirical analysis shows that Trade Openness is significant and negatively influence the flow of in India during the entire period. The dummy variable (financial crisis) has no substantial impact on FDI inflows in India.

Table 6: ARDL Bounds Test

ARDL Bounds Test				
Null Hypothesis: No long-run relationships exist				
Test Statistic	Value	k		
F-statistic	10.13540	5		
Critical Value Bounds				
Significance	I(0) Bound	I(1) Bound		
10%	2.26	3.35		
5%	2.62	3.79		
2.5%	2.96	4.18		
1%	3.41	4.68		

Table 7: Estimated Co-Integrating Form and Long-Run Coefficients using ARDL Approach for FDI and its Determinants

ARDL Cointegrating And Long Run Form				
Dependent Variable: FDI				
Cointegrating Form				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(FDI(-1))	-0.202498	0.062881	-3.220315	0.0014
D(NEER)	19.773136	11.679241	1.693015	0.0917
D(WPI)	18.500441	9.024865	2.049941	0.0414
D(IIP)	10.631784	4.743283	2.241440	0.0259
D(TO)	0.023514	3.090589	0.007608	0.9939
D(TO(-1))	7.703344	3.109982	2.476974	0.0139
D(DM)	126.50896	249.697910	0.506648	0.6128
CointEq(-1)	-0.544750	0.072248	-7.540039	0.0000
Cointeq = FDI - (36.2976*NEER + 33.9613*WPI + 19.5168*IIP -5.0635*TO + 232.2329*DUMMY_VARIABLE -7506.1177)				
Long Run Coefficients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
NEER	36.297601	21.074756	1.722326	0.0862
WPI	33.961311	16.129773	2.105505	0.0362
IIP	19.516796	8.689612	2.245992	0.0256
TO	-5.063451	2.977765	-1.700420	0.0903
DM	232.232851	450.786753	0.515172	0.6069
C	-7506.117684	3296.5612	-2.276954	0.0236

In short, India's Wholesale Price Index, Exchange Rate volatility and the Index of Industrial Production have positively influence the flow of FDI in India and Trade Openness (government policy) is negatively statistically significant for the flow of FDI in India. The short-run nexus between Foreign Direct Investment and the macroeconomic variables was calculated using an Error Correction Model (ECM). The short run coefficient and Error Correction Term of macro-economic variables of Foreign Direct Investment flow is presented in Table 7.

In a short-run analysis Exchange Rate, Wholesale Price Index, Index of Industrial Production and Trade Openness are positive significant relationship of FDI flows in India. The Error Correction Model (ECM) provides a framework for establishing links between the short-run and long-run approaches to econometric modelling. The coefficient of the Error Correction Term (ECT) is highly significant with expected sign, which confirm the result of bound test for co-integration. It is the speed of adjustment towards long run equilibrium. The equilibrium correlation coefficient is estimated -0.54 is highly significant at one percent. If the Error Correction Term is negative in sign and significant, we can say that there is a long run causality running from macro-economic variables to FDI in India. In other wards nearly 54 percent of any disequilibrium between these variables is corrected within one period (One month). The system is getting adjusted towards long run equilibrium at the speed of 54%.

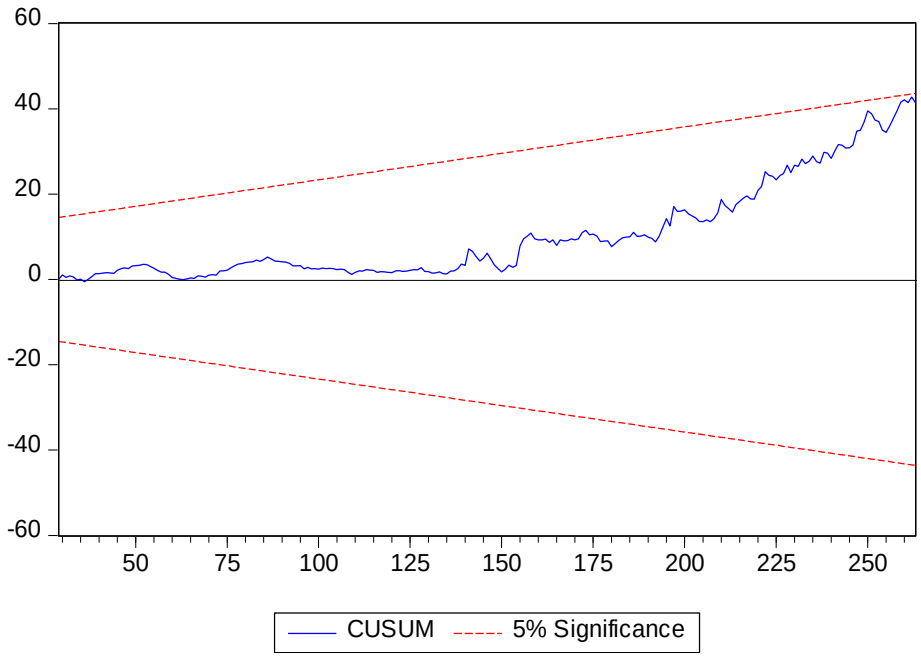
9.2 Diagnostic test or stability test

The CUSUM test (Brown, Durbin, & Evans, 1975) are based on the cumulative sum of the recursive residuals. It helps to show if coefficient of regression are changing systematically or not. Figure 4 provides very clear image that the model is stable. If the blue line lies between or within red lines we accept the null hypothesis i.e. parameters are stable.

10.0 Conclusion

It is generally accepted that Foreign Investment especially Foreign Direct Investment flows are crucial for the development of the host economy. This study attempts to determine the impact of macro-economic variables of FDI flows in Indian economy. On the basis of ARDL approach shows that there exists a long run relationship among the variables of the model. It is found that India's Wholesale Price Index, Exchange Rate Volatility and Index of Industrial Production have positively determined to the flow of FDI in India and Trade Openness (government policy) is negatively significant for the flow of FDI in India.

Figure 4: Cumulative Sum of Recursive Residuals of FDI and Determinants in India



India has continually tried to attract FDI from the world's major investors. With liberalization of Indian economy inflows of FDI has gone up substantially. The average monthly flow of FDI is $1370.29 million. The major contribution of FDI flows is through equity capital. Only the share of reinvested profits and other capital of total FDI inflows are comparatively low. Finally, it can be concluded this study shows that what are the various macro-economic factors for determining FDI flows in the Indian economy. In short there is a long run equilibrium between FDI and five explanatory macro-economic variables in India.

Endnotes

1. Equity capital is the foreign direct investor's purchase of share of an enterprise in a state other than its own.
2. Reinvested earnings comprise the direct investors' share (in proportion to direct equity participation) of earnings not distributed as dividends by the affiliates, or earnings not remitted to the direct investor. Such retained profits by affiliates are reinvested.

3. Other capital or Intra-company loans or intra-company debt transactions refer to short- or long-term borrowing and lending of funds between direct investors (parent enterprises) and affiliate enterprises.

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